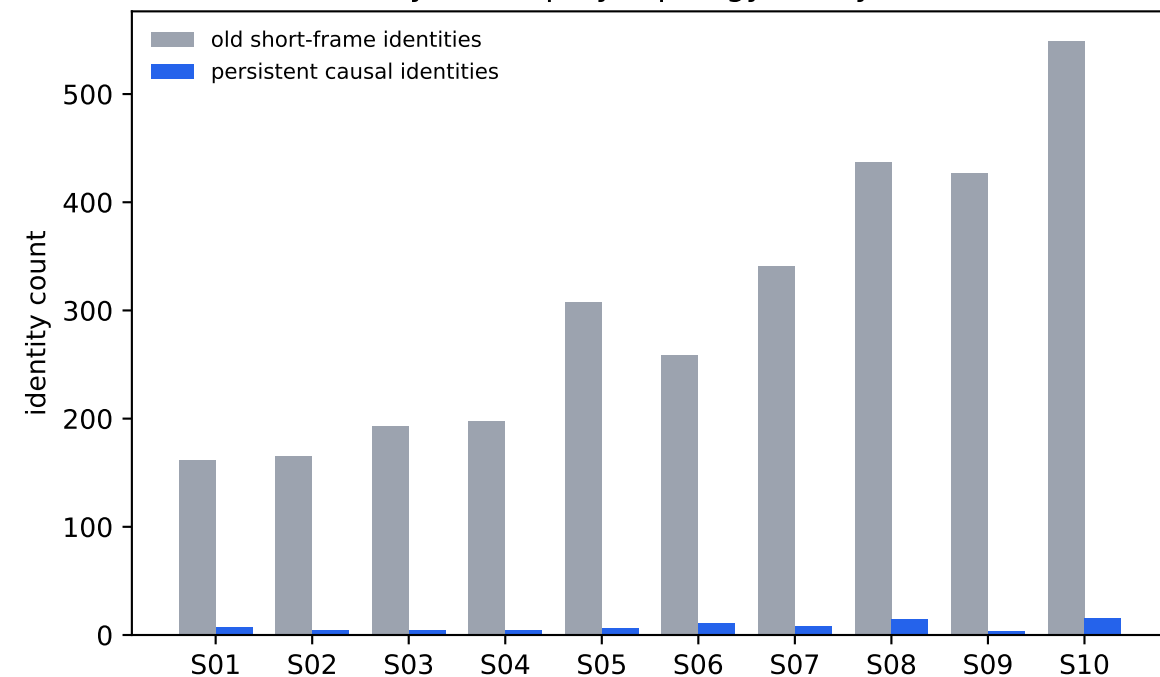
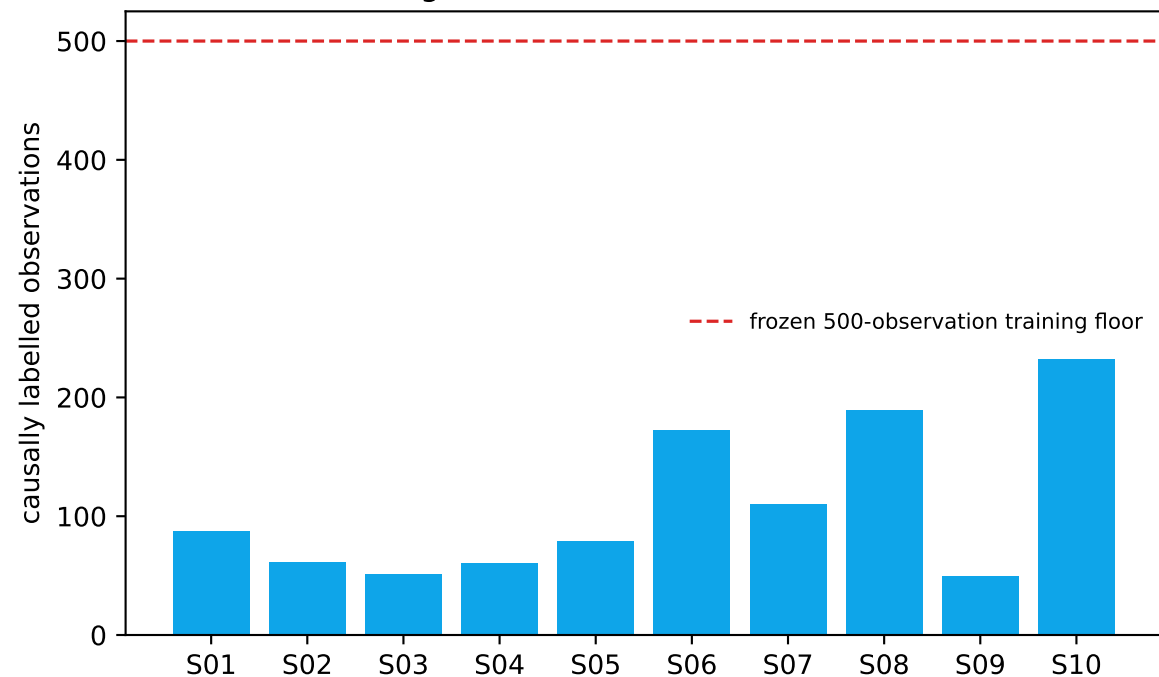


Persistent bidirectional causal geometry change-point Teacher — C01-C08

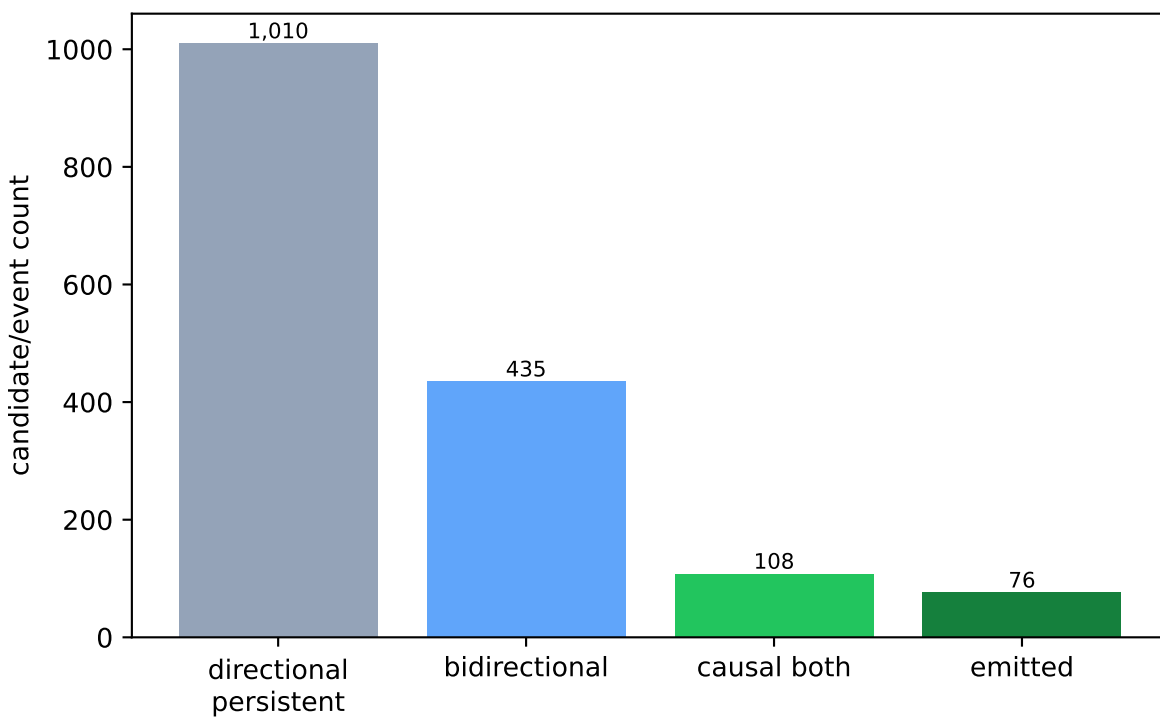
a Teacher identity cleanup by topology family



b Effective training observations



c Fail-closed evidence funnel



d Detection delay after boundary

